Inequalities - Solutions

Level 1

Problem 1. Let a, b, c be nonnegative real numbers satisfying $a+b+c=1$. Prove that

$$a^3+b^3+c^3+6abc \geqslant \frac{1}{4}.$$

Solution. Since $a+b+c=1$, it is the same as proving

$$a^3+b^3+c^3+6abc \geqslant \frac{1}{4}(a+b+c)^3.$$

Multiplying both sides by 4 and expanding, we need to prove

$$3(a^3+b^3+c^3)-3(a^2b+a^2c+b^2a+b^2c+c^2a+c^2b)+18abc \geqslant 0.$$

By Schur's inequality, we have

$$(a^3+b^3+c^3)-(a^2b+a^2c+b^2a+b^2c+c^2a+c^2b)+3abc \geqslant 0.$$

This yields

$$\begin{aligned}
&3(a^3+b^3+c^3)-3(a^2b+a^2c+b^2a+b^2c+c^2a+c^2b)+18abc \\
&= 3[(a^3+b^3+c^3)-(a^2b+a^2c+b^2a+b^2c+c^2a+c^2b)+3abc]+9abc \\
&\geqslant 0.
\end{aligned}$$

Equality holds when $(a,b,c)=\left(\frac{1}{2},\frac{1}{2},0\right), \left(\frac{1}{2},0,\frac{1}{2}\right), \left(0,\frac{1}{2},\frac{1}{2}\right)$.

Problem 2. Let a, b, c be nonnegative real numbers. Prove that

$$a^3+b^3+c^3+ab^2+bc^2+ca^2 \geqslant 2(a^2b+b^2c+c^2a).$$

Solution. By the AM-GM inequality, we have

$$\begin{aligned}
a^3+ab^2 &\geqslant 2\sqrt{(a^3)(ab^2)} = 2a^2b, \\
b^3+bc^2 &\geqslant 2\sqrt{(b^3)(bc^2)} = 2b^2c, \\
c^3+ca^2 &\geqslant 2\sqrt{(c^3)(ca^2)} = 2c^2a.
\end{aligned}$$

Adding these, we obtain the desired inequality. Equality holds when $a=b=c$.

Problem 3. Let x, y, z be positive real numbers. Prove that

$$\frac{x}{y+z} + \frac{y}{z+x} + \frac{z}{x+y} \geqslant \frac{(x+y+z)^2}{2(xy+yz+zx)}.$$

Solution. By the Cauchy-Schwarz inequality, we have

$$[x(y+z) + y(z+x) + z(x+y)]\left(\frac{x}{y+z} + \frac{y}{z+x} + \frac{z}{x+y}\right) \geqslant (x+y+z)^2.$$

Dividing both sides by $2(xy + yz + zx)$, this yields the desired inequality. Equality holds when $x = y = z$.

Problem 4. Let a, b, c be positive real numbers such that $a > c$ and $b > c$. Prove that

$$\sqrt{c(a-c)} + \sqrt{c(b-c)} \leqslant \sqrt{ab}.$$

Solution. By the Cauchy-Schwarz inequality, we have

$$[c + (a-c)][(b-c) + c] \geqslant \left(\sqrt{c(b-c)} + \sqrt{(a-c)c}\right)^2.$$

This is the same as the desired inequality.

Problem 5. Let a, b, c, d be positive real numbers. Prove that

$$(a^2 + b^2 + c^2 + d^2)^2 \geqslant (a+b)(b+c)(c+d)(d+a).$$

Solution. By the AM-GM inequality, we have

$$\begin{aligned}(a+b)(c+d) &= ac + bd + ad + bc \\ &\leqslant \frac{a^2+c^2}{2} + \frac{b^2+d^2}{2} + \frac{a^2+d^2}{2} + \frac{b^2+c^2}{2} \\ &= a^2 + b^2 + c^2 + d^2.\end{aligned}$$

Similarly, we have $(b+c)(d+a) \leqslant a^2 + b^2 + c^2 + d^2$. The result follows by multiplying these inequalities. Equality holds when $a = b = c = d$.

Problem 6. Let x and y be real numbers. Prove that

$$x^2 + y^2 + 2\sqrt{(x-1)^2 + y^2} \geqslant 1.$$

Solution. Consider the points $A(x, y)$, $B(1, 0)$ and $O(0, 0)$ on the Cartesian plane. By the triangle inequality, we have $OA + AB \geqslant OB$. This is the same as

$$\sqrt{x^2 + y^2} + \sqrt{(x-1)^2 + y^2} \geqslant 1. \tag{1}$$

Let $a = \sqrt{x^2 + y^2}$. Then (1) becomes $\sqrt{(x-1)^2 + y^2} \geqslant 1 - a$. Therefore,

$$x^2 + y^2 + 2\sqrt{(x-1)^2 + y^2} \geqslant a^2 + 2(1-a) = (a-1)^2 + 1 \geqslant 1.$$

Equality holds when $(x, y) = (1, 0)$.

Problem 7. Let $n \geqslant 2$ and let $a_1, a_2, \ldots, a_n$ be real numbers greater than 1. Prove that

$$(a_1 + 1)(a_2 + 1) \cdots (a_n + 1) < 2^{n-1}(a_1 a_2 \cdots a_n + 1). \tag{1}$$

Solution. We prove the inequality (1) by induction on n. When $n = 2$, we need to prove

$$(a_1 + 1)(a_2 + 1) < 2(a_1 a_2 + 1)$$

for any $a_1, a_2 > 1$. This can be rewritten as $(a_1 - 1)(a_2 - 1) > 0$, which is obviously true.

Assume (1) holds for some $n = k \geqslant 2$. When $n = k + 1$, we need to prove

$$(a_1 + 1)(a_2 + 1) \cdots (a_k + 1)(a_{k+1} + 1) < 2^k (a_1 a_2 \cdots a_{k+1} + 1). \tag{2}$$

Applying the inductive hypothesis to the first k terms, we have

$$(a_1 + 1)(a_2 + 1) \cdots (a_k + 1)(a_{k+1} + 1) < 2^{k-1}(a_1 a_2 \cdots a_k + 1)(a_{k+1} + 1).$$

By letting $b = a_1 a_2 \cdots a_k$, it remains to prove

$$(b + 1)(a_{k+1} + 1) < 2(b a_{k+1} + 1).$$

This is the same as the base case $n = 2$ since $b, a_{k+1} > 1$. Thus, we have proven (2). By induction, the desired inequality holds for any integer $n \geqslant 2$.

Problem 8. Given three distinct nonzero real numbers, prove that we can label them as a, b, c in some order such that

$$\frac{a}{b} + \frac{b}{c} > \frac{a}{c} + \frac{c}{a}.$$

Solution. Let the real numbers be x, y, z. Suppose on the contrary that

$$\frac{a}{b} + \frac{b}{c} \leqslant \frac{a}{c} + \frac{c}{a}$$

for any permutation a, b, c of x, y, z. There are 6 permutations in total. Adding up these 6 inequalities, both sides are the same since $\sum_{sym} \frac{a}{b} = \sum_{sym} \frac{b}{c} = \sum_{sym} \frac{a}{c} = \sum_{sym} \frac{c}{a}$. Therefore, all the inequalities must be equalities. In particular, we have

$$\begin{aligned}
\frac{x}{y} + \frac{y}{z} &= \frac{x}{z} + \frac{z}{x}, \\
\frac{y}{z} + \frac{z}{x} &= \frac{y}{x} + \frac{x}{y}, \\
\frac{z}{x} + \frac{x}{y} &= \frac{z}{y} + \frac{y}{z}.
\end{aligned}$$

Adding these and cancelling like terms, we obtain

$$\frac{x}{y} + \frac{y}{z} + \frac{z}{x} = \frac{x}{z} + \frac{y}{x} + \frac{z}{y}.$$

Clearing denominators, this becomes $x^2z + y^2x + z^2y = x^2y + y^2z + z^2x$, i.e.

$$(x - y)(y - z)(z - x) = 0.$$

This contradicts the fact that x, y, z are distinct. So the assertion must hold.

Level 2

Problem 9. (JBMO shortlist 2002) Let a, b, c be positive real numbers. Prove the inequality:

$$\frac{a^3}{b^2} + \frac{b^3}{c^2} + \frac{c^3}{a^2} \geqslant \frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{a}.$$

Solution. By the Cauchy-Schwarz inequality, we have

$$(a + b + c)\Big(\frac{a^3}{b^2} + \frac{b^3}{c^2} + \frac{c^3}{a^2}\Big) \geqslant \Big(\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{a}\Big)^2$$

and

$$(b + c + a)\Big(\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{a}\Big) \geqslant (a + b + c)^2.$$

Multiplying these inequalities and removing some common factors, we obtain the desired inequality. Equality holds when $a = b = c$.

Problem 10. Let x and y be nonnegative real numbers. Prove that

$$\frac{1}{(1+x)^2}+\frac{1}{(1+y)^2}\geqslant\frac{1}{1+xy}.$$

Solution. By the Cauchy-Schwarz inequality, we have

$$\frac{(1+xy)(x+y)}{y}=(1+xy)\Big(1+\frac{x}{y}\Big)\geqslant(1+x)^2,$$
$$\frac{(1+xy)(x+y)}{x}=(1+xy)\Big(1+\frac{y}{x}\Big)\geqslant(1+y)^2.$$

This yields

$$\frac{1}{(1+x)^2}+\frac{1}{(1+y)^2}\geqslant\frac{y}{(1+xy)(x+y)}+\frac{x}{(1+xy)(x+y)}=\frac{1}{1+xy}.$$

Equality holds when $x=y=1$.

Problem 11. Let a,b,c,d be positive real numbers such that $abcd=1$. Prove that

$$\frac{1+ab}{1+a}+\frac{1+bc}{1+b}+\frac{1+cd}{1+c}+\frac{1+da}{1+d}\geqslant 4.$$

Solution. Let $a=\frac{x}{w}$, $b=\frac{y}{x}$, $c=\frac{z}{y}$ and $d=\frac{w}{z}$. The inequality becomes

$$\frac{w+y}{w+x}+\frac{x+z}{x+y}+\frac{y+w}{y+z}+\frac{z+x}{z+w}\geqslant 4.$$

Note that

$$\frac{w+y}{w+x}+\frac{y+w}{y+z}=(w+y)\Big(\frac{1}{w+x}+\frac{1}{y+z}\Big)\geqslant(w+y)\times\frac{4}{w+x+y+z}$$

by the AM-HM inequality. Similarly, we have

$$\frac{x+z}{x+y}+\frac{z+x}{z+w}\geqslant(x+z)\times\frac{4}{w+x+y+z}.$$

Thus,

$$\frac{w+y}{w+x}+\frac{x+z}{x+y}+\frac{y+w}{y+z}+\frac{z+x}{z+w}\geqslant\frac{4(w+y)}{w+x+y+z}+\frac{4(x+z)}{w+x+y+z}=4.$$

Equality holds when $x=z$ and $w=y$, i.e. $a=c=\frac{1}{b}=\frac{1}{d}$.

Problem 12. Let x, y, z be nonnegative real numbers. Prove that

$$x^2 + y^2 + z^2 + 2xyz + 1 \geqslant 2(xy + yz + zx).$$

Solution 1. Putting everything to the left, we consider

$$x^2 + y^2 + z^2 + 2xyz + 1 - 2(xy + yz + zx) = (x-1)^2 + (y-z)^2 + 2x(y-1)(z-1).$$

Note that two of $x-1$, $y-1$ and $z-1$ are of the same sign. WLOG assume $(y-1)(z-1) \geqslant 0$. Then it is obvious that the above expression is nonnegative. Equality holds when $x = y = z = 1$.

Solution 2. Rewrite the desired inequality as

$$x^2 + 2(yz - y - z)x + (y^2 - 2yz + z^2 + 1) \geqslant 0.$$

Note that $y^2 - 2yz + z^2 + 1 = (y-z)^2 + 1 > 0$. This is obviously true if $yz - y - z \geqslant 0$. In the following, we assume $yz - y - z < 0$. It suffices to prove the discriminant is non-positive, i.e.

$$4(yz - y - z)^2 - 4(y^2 - 2yz + z^2 + 1) \leqslant 0. \tag{1}$$

After expansion and re-grouping the terms, (1) becomes

$$yz(y-2)(z-2) \leqslant 1. \tag{2}$$

If $y \geqslant 2$, then $z \leqslant 2$ since otherwise $(y-1)(z-1) \geqslant 1$, which contradicts $yz - y - z < 0$. In this case (2) trivially holds. Similarly, if $z \geqslant 2$, then $y \leqslant 2$ and (2) still holds. It remains to consider the case $y, z < 2$. We rewrite (2) as

$$z(2-z)y^2 - 2z(2-z)y + 1 \geqslant 0.$$

Since $z < 2$, this inequality holds if $\Delta = 4z^2(2-z)^2 - 4z(2-z) \leqslant 0$. This can be factorized as $\Delta = -4z(2-z)(z-1)^2$, which must be $\leqslant 0$ as $z < 2$. The proof is complete. Equality holds when $x = y = z = 1$.

Problem 13. Let $x_1, x_2, \ldots, x_n$ be nonnegative real numbers satisfying

$$x_1 + x_2 + \cdots + x_n \leqslant \frac{1}{2}.$$

Prove that

$$(1 - x_1)(1 - x_2) \cdots (1 - x_n) \geqslant \frac{1}{2}.$$

Solution. We shall prove by induction that $x_1 + x_2 + \cdots + x_n \leqslant \frac{1}{2}$ implies

$$(1 - x_1)(1 - x_2) \cdots (1 - x_n) \geqslant 1 - (x_1 + x_2 + \cdots + x_n). \tag{1}$$

When $n = 1$, the inequality is indeed an equality. Assume (1) holds for some integer $n = k \geqslant 1$. When $n = k + 1$, we need to prove $x_1 + x_2 + \cdots + x_{k+1} \leqslant \dfrac{1}{2}$ implies

$$(1 - x_1)(1 - x_2) \cdots (1 - x_k)(1 - x_{k+1}) \geqslant 1 - (x_1 + x_2 + \cdots + x_{k+1}). \tag{2}$$

Since the variables are nonnegative, the condition implies $x_1 + x_2 + \cdots + x_k \leqslant \dfrac{1}{2}$ and $x_{k+1} < 1$. Applying the inductive hypothesis to the first k terms of (2), we have

$$(1 - x_1)(1 - x_2) \cdots (1 - x_k)(1 - x_{k+1}) \geqslant [1 - (x_1 + x_2 + \cdots + x_k)](1 - x_{k+1}).$$

By letting $y = x_1 + x_2 + \cdots + x_k$, it remains to prove

$$(1 - y)(1 - x_{k+1}) \geqslant 1 - (y + x_{k+1}).$$

After simplification, this is simply $yx_{k+1} \geqslant 0$, which is obviously true. This proves (2), hence (1) holds by induction. It now follows easily that

$$(1 - x_1)(1 - x_2) \cdots (1 - x_n) \geqslant 1 - (x_1 + x_2 + \cdots + x_n) \geqslant \frac{1}{2}.$$

Equality holds when one of $x_1, x_2, \ldots, x_n$ is $\dfrac{1}{2}$ and the others are 0.

Problem 14. (CSMO 2019 Grade 10 Problem 1) Find the largest real number k, such that for any positive real numbers a, b,

$$(a + b)(ab + 1)(b + 1) \geqslant kab^2. \tag{1}$$

Solution. By the AM-GM inequality, we have

$$\begin{aligned} a + \frac{b}{2} + \frac{b}{2} &\geqslant 3\sqrt[3]{\frac{ab^2}{4}}, \\ \frac{ab}{2} + \frac{ab}{2} + 1 &\geqslant 3\sqrt[3]{\frac{a^2b^2}{4}}, \\ \frac{b}{2} + \frac{b}{2} + 1 &\geqslant 3\sqrt[3]{\frac{b^2}{4}}. \end{aligned}$$

Multiplying these inequalities, we get

$$(a + b)(ab + 1)(b + 1) \geqslant \frac{27}{4}ab^2.$$

Therefore, the answer is at least $\dfrac{27}{4}$. By considering the equality case, we take $a = 1$ and $b = 2$. The inequality (1) becomes $27 \geqslant 4k$. So we need $k \leqslant \dfrac{27}{4}$. It follows that the largest k is $\dfrac{27}{4}$.

Problem 15. (JBMO 2002 Problem 4) Prove that for all positive real numbers a, b, c the following inequality takes place

$$\frac{1}{b(a+b)} + \frac{1}{c(b+c)} + \frac{1}{a(c+a)} \geqslant \frac{27}{2(a+b+c)^2}.$$

Solution. By Hölder's inequality, we have

$$\begin{aligned}&(b+c+a)[(a+b)+(b+c)+(c+a)]\Big(\frac{1}{b(a+b)} + \frac{1}{c(b+c)} + \frac{1}{a(c+a)}\Big)\\ \geqslant\ &(1+1+1)^3 = 27.\end{aligned}$$

Dividing both sides by $2(a+b+c)^2$, this yields the desired inequality. Equality holds when $a = b = c$.

Problem 16. (Saudi Arabia TST 2019) Let n be a positive integer and let $a_1, a_2, \dots, a_n$ be any real numbers. Prove that there exist $m, k \in \{1, 2, \dots, n\}$ such that

$$\left|\sum_{i=1}^{m} a_i - \sum_{i=m+1}^{n} a_i\right| \leqslant |a_k|.$$

Solution. Let $M = \max\limits_{1 \leqslant i \leqslant n} \{|a_i|\}$ and let $s_i = a_1 + a_2 + \cdots + a_i$ for $i = 1, 2, \dots, n$. Note that

$$\sum_{i=1}^{m} a_i - \sum_{i=m+1}^{n} a_i = s_m - (s_n - s_m) = 2s_m - s_n.$$

Suppose on the contrary that the assertion does not hold. This means

$$|2s_m - s_n| > M \tag{1}$$

for each $m = 1, 2, \dots, n$. If we define $s_0 = 0$, then this also holds for $m = 0$ since

$$|0 - s_n| = |s_n| = |2s_n - s_n| > M.$$

WLOG we may assume $s_n \geqslant 0$ (by flipping the signs of all a_i if necessary). Then $2s_0 - s_n = -s_n \leqslant 0$ and $2s_n - s_n = s_n \geqslant 0$. By the assumption (1), we must have $2s_0 - s_n < -M$ and $2s_n - s_n > M$. Thus, there exists an index $0 \leqslant j \leqslant n-1$ such that $2s_j - s_n < -M$ and $2s_{j+1} - s_n > M$. Then we have

$$2M < (2s_{j+1} - s_n) - (2s_j - s_n) = 2s_{j+1} - 2s_j = 2a_{j+1}.$$

This yields $M < a_{j+1}$, which contradicts the definition of M. Therefore, the desired inequality must hold for some m, k.